Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/29/2012
PI Name:	Sowa, Marianne Ph.D.		
Project Title:	Integrated experimental and computational study of radiation induced matrix remodeling in a human skin equivalent		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marianne.sowa@pnl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	509-371-6898
Organization Name:	Battelle Pacific Northwest National Laboratory		
PI Address 1:	Biological Sciences Division		
PI Address 2:	PO BOX 999, MS J4-02		
PI Web Page:			
City:	Richland	**State:**	WA
Zip Code:	99352-0999	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	11/01/2009	**End Date:**	04/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/30/2014 per NSSC information (Ed., 4/5/2013) NOTE: End date changed to 4/30/2013, per NSSC information (Ed., 8/30/2012) NOTE: Period of performance changed to 11/1/09-10/31/12 (from 10/9/09-10/8/12) per NSSC info (3/2010) NOTE: Period of performance changed to 10/9/09-10/8/12 (from 11/15/09-11/14/12) per NSSC info (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shankaran, Harish (Battelle, Pacific Northwest Divsion)		
Grant/Contract No.:	NNX10AB06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current risk estimates for manned space flight are based predominantly on human epidemiology data following low linear energy (LET) radiation exposures. These data are scaled to the radiation types and fluences commonly encountered in space through the application of radiation quality factors and dose, dose rate modification factors. NASA and the scientific community acknowledge the limitations of this approach, however appropriate data on the radiation quality dependence for high energy, high atomic number ions (HZE) such as those found in space are currently unavailable making for large uncertainties in risk estimates. This application proposes an integrated experimental and computational strategy to develop a predictive understanding of the link between cell signaling, gene expression and proximal tissue responses to radiations commonly found in space. We will achieve this by quantitatively measuring the appropriate variables that represent these processes and using the experimental data to construct a mathematical model of the relationships between the measured variables. The constructed model will provide us with a mechanistic understanding of the effect of radiation quality on tissue responses. Experiments are specifically designed to obtain information on how track structure modifies the radiation induced signaling pathways important to maintaining tissue homeostasis. Using a 3D skin tissue model, we will determine changes in gene expression and protein activation levels for targeted matrix signaling components as a function of radiation quality for two constant beam fluences. Ion beams to be used are: 600 MeV/u iron, 400 MeV/u silicon and 250 Mev/u oxygen. We will use immunohistochemistry to monitor overall matrix integrity as well as cell proliferation and apoptosis rates. We will use the quantitative data collected to determine the parameters of a coarse-grained model involving the key modulators of matrix remodeling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will provide fundamental mechanistic information on the responses of skin to radiation.
Task Progress:	This is the third year of funding for grant NNX10AB06G. The goal of this project is to use a combined experimental and computational approach to develop a predictive understanding of the link between cell signaling, gene expression, and proximal tissue responses to radiations commonly encountered in space. We hypothesized that radiation quality differentially affects the tissue response by altering signaling pathways and cellular responses. We have characterized molecular, cellular, and tissue level changes induced by radiation – an essential step to construct an integrated understanding of the response. This enabled us to identify the critical biological endpoints for further investigation. We have observed radiation quality-dependent effects on endpoints related to the cancer development processes (proliferation, differentiation, extracellular matrix remodeling). For some of these endpoints, we have constructed mathematical models to integrate information, and to link cell-level responses to changes in tissue phenotype. All experiments were performed with a commercially available three dimensional (3D) skin equivalent composed of normal human epidermal keratinocytes and dermal fibroblasts. This is the most robust experimental model available for mechanistic studies of radiation effects on epithelial tissues. To study radiation quality dependent effects we have exposed the skin model to low linear energy transfer (LET) radiation X-rays, and a range of high atomic number ions (HZE). We participated in all three runs in 2011 with a total of ~37 staff days at NSRL. The final NSRL exposure for the current funding cycle will occur at the fall 2012 run. All exposures were conducted with constant ion fluences (3.6 x 104 and 1.1 x 105 particles/cm2). By following this irradiation protocol, we can focus on the delta ray contribution to the induced tissue responses. To date, we have exposed ~1000 skin samples, and have generated histological slides, total mRNA, and protein for investigating the tissue response. Our data indicates that HZE radiation induces LET, and fluence dependent changes in the fraction of proliferating cells in the tissue. We also observed marked changes in the thickness of the epidermal layer, indicative of altered cell proliferation and/or differentiation in the tissue. To understand how specific changes in cell behavior could result in these observations, we have collected additional data on the tissue differentiation profile using keratin-10 (labels all differentiated cells in the spinous and granular layers) and filaggrin (labels the granular layer) immunostaining. We have analyzed the results using a mathematical model that links cell behavior changes to the experimentally measured tissue-level quantities. Data on mRNA expression, signaling and metalloprotease activities viewed together in the context of an ECM remodeling pathway map indicated that connective tissue growth factor (CTGF) and plasmin were important drivers of ECM changes in the system. We have validated the role of the plasminogen system by independently altering its activity. For HZE ions we have observed differential effects on CTGF intracellular localization in the epidermal and dermal layers as a function of LET. With regard to other ECM-related genes, using an mRNA array for ECM remodeling with 84 genes, we found strong LET-dependence in the response. While the ECM pathway was found to be affected both at high and low LET, there was little overlap in the specific transcriptional targets. Interestingly, high LET exposures exhibited a very different cell type specific response than X-rays where dermal responses predominated. HZE exposures resulted in almost a complete absence of transcriptional changes in the dermis. This is likely due to the inhomogeneity of the energy deposition pattern for high LET radiation, and the low density of cells (fibroblasts) in the dermal layer (in our skin model the cell density is ~85% lower in the dermis compared to the epidermis). We have made significant progress towards our objective of understanding the tissue response to radiations commonly encountered in space. This grant has contributed to three publications; three additional manuscripts are in preparation. In the current proposal, we will further build on this work to understand the mechanisms underlying the radiation quality dependence of cell proliferation, differentiation and ECM remodeling in this tissue model.
Bibliography Type:	Description: (Last Updated: 09/23/2019)
Articles in Peer-reviewed Journals	Waters KM, Stenoien DL, von Neubeck C, Sowa MB, Chrisler WB, Tan R, Sontag RL, Weber TJ. "Annexin A2 Modulates Radiation-Sensitive Transcriptional Programming and Cell Fate." Radiation Research. In press as of August 2012. , Aug-2012
Articles in Peer-reviewed Journals	von Neubeck C, Shankaran H, Karin NJ, Kauer PM, Chrisler WB, Wang X, Robinson RJ, Waters KM, Tilton SC, Sowa MB. "Cell type-dependent gene transcription profile in a three-dimensional human skin tissue model exposed to low doses of ionizing radiation: implications for medical exposures." Environ Mol Mutagen. 2012 May;53(4):247-59. Epub 2012 Feb 20. http://dx.doi.org/10.1002/em.21682 ; PubMed PMID: 22351304 , May-2012

Articles in Peer-reviewed Journals	Shankaran H, Weber TJ, von Neubeck C, Sowa MB. "Using imaging methods to interrogate radiation-induced cell signaling." Radiat Res. 2012 Apr;177(4):496-507. Epub 2012 Mar 1. Review. PubMed PMID: 22380462 , Apr-2012